3D Matrices: Diagonalisation

Questions 1–10

Question 1. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 0 & -4 \\ 0 & 2 & -5 \\ 0 & 0 & -3 \end{pmatrix}$. Two of the eigenvalues of $\mathbf{A}$ are $\lambda_1 = -3$ and $\lambda_2 = 1$ with eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ respectively. A matrix $\mathbf{D}$ is diagonal and is defined such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$.

a) Find the value of the third eigenvalue λ_3.

b) By finding the eigenvector corresponding to λ_3, determine a possible matrix $\mathbf{P}$ and hence $\mathbf{P}^{-1}$.

c) By finding $\mathbf{P}^{-1}\mathbf{AP}$, find a possible matrix $\mathbf{D}$.

Question 2. For each of the following, by finding eigenvalues $\lambda_1 < \lambda_2 < \lambda_3$ and their respective eigenvectors to find a matrix $\mathbf{P}$, and using your calculator to find $\mathbf{P}^{-1}$, determine diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$:

a) $\mathbf{A} = \begin{pmatrix} 5 & 1 & -5 \\ 2 & 3 & -4 \\ 2 & 1 & -2 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 4 & 3 \end{pmatrix}$

Question 3. Given the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$:

a) Find the eigenvalues and corresponding eigenvectors of matrix $\mathbf{A}$.

b) Explain why this matrix cannot be diagonalised.

Question 4. Find each of the following:

a) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^6$

b) $\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix}^9$

c) $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^k$ for constants a, b, c and k.

Question 5. The matrix $\mathbf{A}$ has eigenvalues $-1, 1, 3$ with corresponding eigenvectors $\begin{pmatrix}1\\0\\1\end{pmatrix}$, $\begin{pmatrix}2\\1\\-1\end{pmatrix}$ and $\begin{pmatrix}0\\1\\3\end{pmatrix}$ respectively. Determine the matrix $\mathbf{A}^4$.

Question 6. The matrix $\mathbf{M}$ has eigenvalues $-5, -1, 1$ with corresponding eigenvectors $\begin{pmatrix}0\\2\\1\end{pmatrix}$, $\begin{pmatrix}1\\1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\0\\3\end{pmatrix}$ respectively. Show that $\mathbf{M}^{10} = \begin{pmatrix} 1 & 0 & 0 \\ -6\times 5^9 + \frac{6}{5} & 4\times 5^9 + \frac{1}{5} & 2\times 5^9 - \frac{2}{5} \\ -3\times 5^9 + \frac{3}{5} & 2\times 5^9 - \frac{2}{5} & 5^9 + \frac{4}{5} \end{pmatrix}$.

Question 7. Given that the matrix $\mathbf{M}$ has eigenvalues $\lambda_1 = \frac{1}{2}$, $\lambda_2 = \frac{1}{3}$ and $\lambda_3 = 1$ with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$:

a) Find $\mathbf{M}^n$ giving your answer in terms of n.

b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

Question 8. Given that the matrix $\mathbf{M}$ has eigenvalues $\lambda_1 = 1, \lambda_2 = \frac{3}{4}$ and $\lambda_3 = \frac{1}{5}$ with eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$:

a) Find $\mathbf{M}^n$ giving your answer in terms of n.

b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

Question 9. Given the matrix $\mathbf{A} = \begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix}$, where k is a non-zero constant. Show that $\mathbf{A}^n = k^n \begin{pmatrix} 2^{n-1} & 0 & 2^{n-1} \\ 0 & 1 & 0 \\ 2^{n-1} & 0 & 2^{n-1} \end{pmatrix}$.

Question 10. Given the matrix $\mathbf{A} = \begin{pmatrix} a - \frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix}$, where a and b are non-zero positive constants and $a < b$:

a) Find the eigenvalues of $\mathbf{A}$ where $\lambda_1 < \lambda_2 < \lambda_3$.

b) Hence, show that $\mathbf{A}^n = \frac{1}{2} \begin{pmatrix} 2a^n - b^n & b^n & b^n - 2a^n \\ 2a^n - 2b^n & 2b^n & 2b^n - 2a^n \\ -b^n & b^n & b^n \end{pmatrix}$.